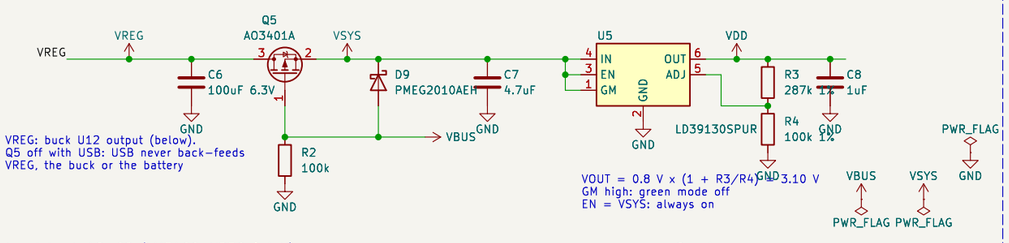
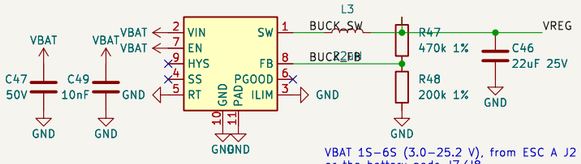
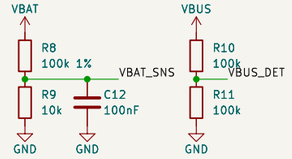
A — Input buck U12 (1S–6S –> VREG 4.1 V), USB/battery power path, LDO



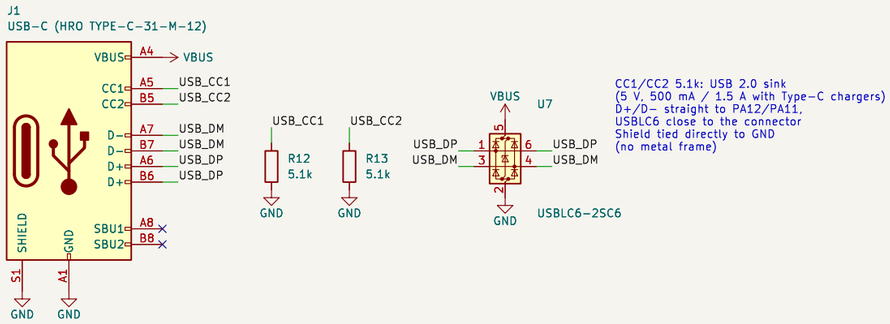
Input buck 1S–6S (LM5166, VREG 4.1 V)



Sensing

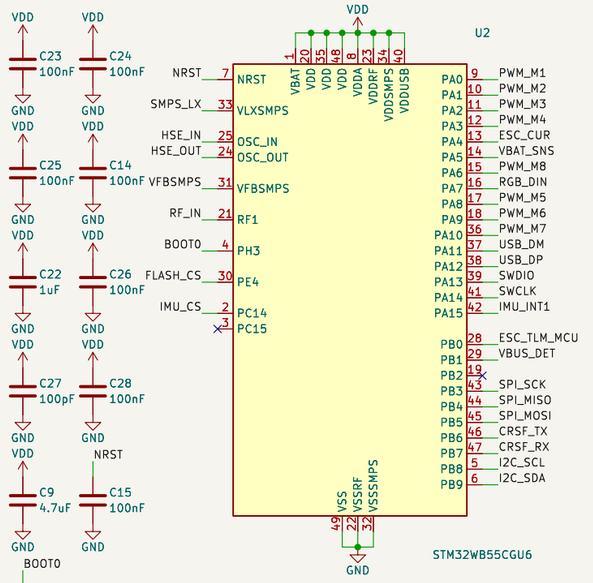


B — USB-C (DFU/CDC, bench supply; no charger), ESD protection

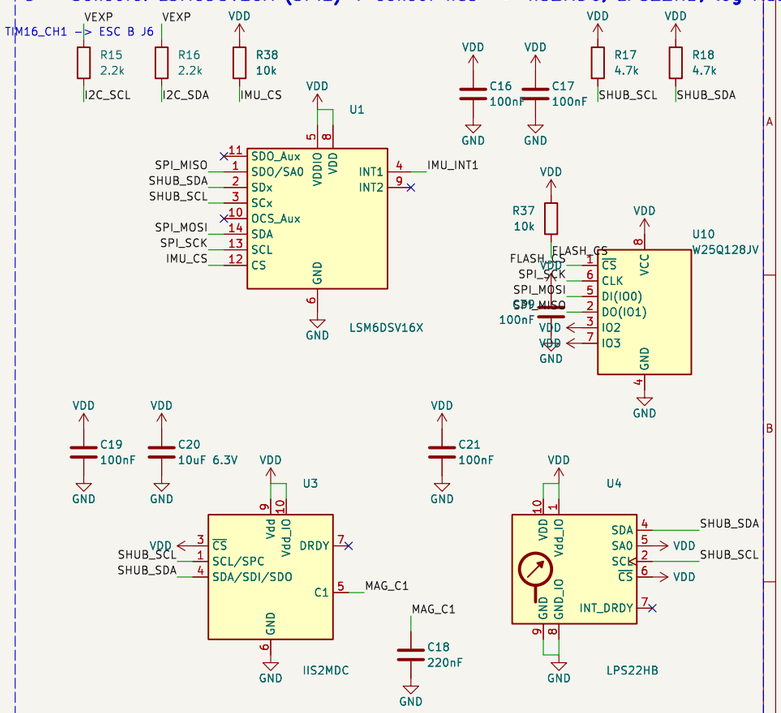


C — STM32WB55CGU6 (QFN48), RF front end, clocks

Pin map (IEKF firmware), rev A:
Motors M1–M4 PA0–PA3 TIM2_CH1–CH4 –> ESC A J2; M5–M7 PA8–PA10 TIM1_CH1–CH3, M8 PA6 TIM16_CH1 –> ESC B J6
(PWM 32 kHz brushed / DShot, bidirectional DShot: every motor timer has DMA)
ADC: PA4 ESC_CUR (IN9), PA5 VBAT_SNS (IN10, x10/110); ESC A serial telemetry PB0 (optional)
SPI1 PB3 SCK / PB4 MISO / PB5 MOSI: IMU (CS PC14) + log flash (CS PE4), IMU_INT1 PA15
USART1 PB6 TX / PB7 RX –> ELRS (CRSF 420 kbaud), J5
I2C1 PB8/PB9 –> expansion bus (Qwiic J3 + J4: GPS board, flow board, sensors)
RGB LEDs PA7 (TIM17_CH1 + DMA, 4 x WS2812B); VBUS_DET PB1: PB2 free
No LSE: RF wake-up HSE/1024, RTC HSE/32. USB PA11/PA12 (DFU, CDC)
SWD PA13/PA14, BOOT0 = PH3 (pull-down 10k)



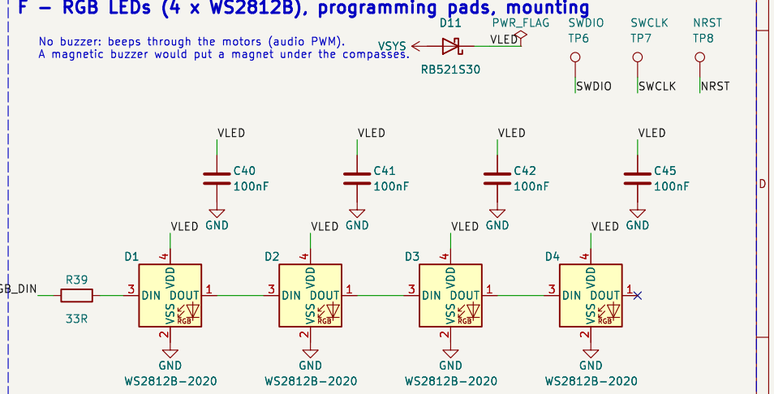
D — Sensors: LSM6DSV16X (SPI1) + sensor hub –> IIS2MDC, LPS22HB; log flash



Topology = IKS4A1 Mode 3: the IEKF firmware
reads the IIS2MDC through the LSM6DSV16X
sensor hub (samples aligned in the FIFO).
Rev E: the IMU host interface is SPI; DRDY/INT2/baro INT not used (FIFO + timestamps)
Addresses on the hub bus: IIS2MDC 0x1E, LPS22HB 0x5D

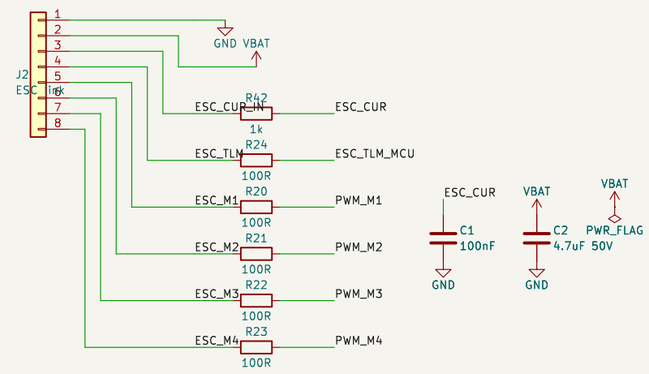
F — RGB LEDs (4 x WS2812B), programming pads, mounting

No buzzer: beeps through the motors (audio PWM).
A magnetic buzzer would put a magnet under the compasses.



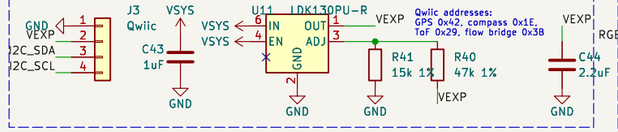
J5: ELRS receiver port,
JST SH 4P

E — ESC link: JST SH 8-pin, common 4-in-1 ESC pinout

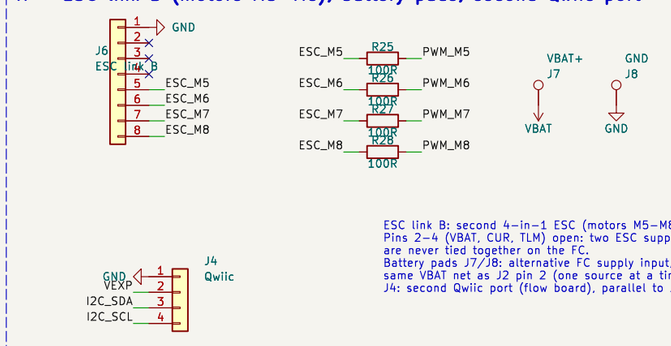


ESC link A (rev A):
- ACwing 1S power board (brushed): CUR = 0.1 V/A.
- Brushless 4-in-1 ESC, 2S–6S: CUR scale set in firmware (about 12 mV/A); telemetry via bidirectional DShot / EDT.
- VBAT 1S–6S feeds the FC (buck U12, block A).
- No charging through the cable (charger on the 1S board).

G — Expansion bus (Qwiic J3, J4) and ELRS port



H — ESC link B (motors M5–M8), battery pads, second Qwiic port



ESC link B: second 4-in-1 ESC (motors M5–M8).
Pins 2–4 (VBAT, CUR, TLM) open: two ESC supplies
are never tied together on the FC.
Battery pads J7/J8: alternative FC supply input,
same VBAT net as J2 pin 2 (one source at a time).
J4: second Qwiic port (flow board), parallel to J3.

Generated automatically: run ERC before release
Pin map: ACwing_FC_A_pinmap.md
Power and motors from a 4-in-1 ESC (JST SH 8-pin ESC A J2, ESC B J6) or the ACwing 1S power b
Outline: frame C (40 x 40, as draft E); holes 20 x 20 (mechanical handoff, update 2026-10-18)

ACwing — Andrea Cirillo

Sheet: /
File: ACwing_FC_A.kicad_sch

Title: ACwing FC rev A — 1S–6S flight controller (STM32WB55CGU6), 4-in-1

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1